

# Dell PowerEdge R510

## Legacy Server Hardware Inspection & Component Documentation

Physical inspection, component documentation, POST/BIOS findings, and visual fault assessment



Figure 1. Dell PowerEdge R510 at the start of the inspection.

### Purpose

This report documents a hands-on inspection of a legacy Dell PowerEdge R510. The goal was to identify installed hardware, record visible physical conditions, capture POST and BIOS information, and document confirmed startup faults. Limited troubleshooting was performed only to restore enough functionality for documentation.

| Result            | Current status                                   |
|-------------------|--------------------------------------------------|
| Platform          | Powers on and reaches POST/BIOS                  |
| Service readiness | Not verified for production use                  |
| Primary concern   | PERC H800 firmware / SDRAM-related startup fault |
| Boot status       | No valid bootable device detected                |

# 1. Inspection Method

The inspection focused on documenting physical configuration and startup behavior while minimizing unnecessary changes.

1. Initial power-on: the server powered up but produced no video.
2. Power was disconnected and the chassis was opened for inspection.
3. Major components, memory banks, storage, RAID hardware, cooling, power supplies, expansion cards, and visible connections were inspected.
4. Both CPU memory banks were initially empty; three compatible 4 GB ECC DDR3 modules were installed in the CPU1 bank so the system could proceed through POST.
5. A loose PERC H700 memory/cache module near the internal drives was carefully reseated.
6. A second RAID memory/cache module appeared mechanically stressed and was intentionally left untouched.
7. After power was restored, video output returned and POST/BIOS information was documented.

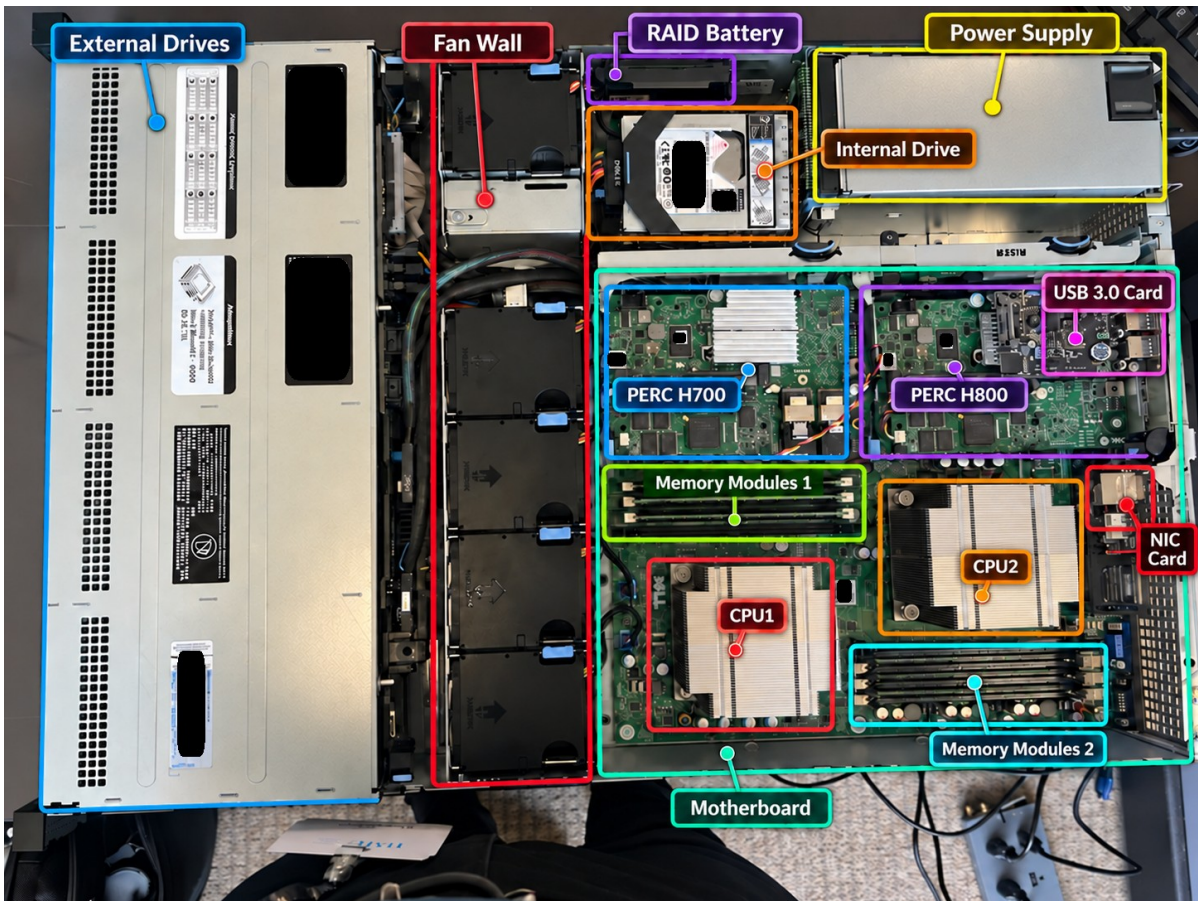


Figure 2. Internal component map used during the inspection.

**Physical condition:** Moderate dust and normal wear were present. No obvious burn damage or visibly swollen motherboard capacitors were observed.

## 2. Hardware Inventory

| Area              | Finding                                                 | Evidence               |
|-------------------|---------------------------------------------------------|------------------------|
| Platform          | Dell PowerEdge R510, 2U rack server                     | Chassis / BIOS         |
| Processors        | 2 x Intel Xeon E5620 @ 2.40 GHz                         | POST / BIOS            |
| Memory            | 3 x 4 GB ECC DDR3, 12 GB total @ 1067 MHz               | Physical / POST / BIOS |
| Front storage     | 4 x Dell-labeled Seagate 146 GB 15K SAS hot-swap drives | Drive labels           |
| Internal storage  | 2 x Western Digital Red WD10JFCX 1 TB SATA drives       | Drive labels           |
| RAID controller 1 | Dell PERC H700 Integrated, DP/N 0R374M                  | Controller label       |
| RAID controller 2 | Dell PERC H800, firmware 12.10.2-0004                   | POST                   |
| Power             | 2 x Dell/Delta 750 W hot-swap power supplies            | Physical labels        |

### Additional Hardware

The server also includes a Dell FR463 3.7 V, 7 Wh RAID battery, Broadcom 5716 dual-port Gigabit Ethernet, a USB 3.0 expansion card, a fan wall, and integrated rear I/O interfaces.

### Memory Configuration

At the beginning of the inspection, neither CPU1 nor CPU2 had local memory installed. Three 4 GB ECC DDR3 modules were installed in the CPU1 bank to enable startup testing. CPU2 remained without local memory and later generated an unsupported-memory warning during POST.

### Storage Capacity

The label-reported raw capacity is approximately 2.584 TB decimal: four 146 GB SAS drives plus two 1 TB SATA drives. This is raw installed capacity only and does not represent usable RAID capacity. RAID level, virtual-disk state, filesystem condition, and data integrity were not verified.

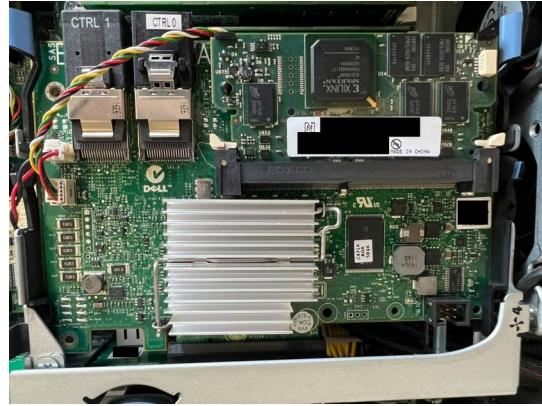
### 3. Storage and RAID Documentation

#### Front SAS Hot-Swap Drives



Four Dell-labeled Seagate Cheetah 15K.7 drives were documented. Each is marked SAS, 146 GB, 15K RPM, model ST3300657SS-H, firmware EH02. The drives were returned to their original bays after documentation.

#### Dell PERC H700 Integrated



The H700 Integrated was identified by physical label. Its memory/cache module was found loose or not fully seated and was carefully reseated before the second startup.

#### Dell PERC H800

The second RAID adapter was identified during POST as a Dell PERC H800. Its memory/cache module appeared visibly arched or mechanically stressed. Because the module appeared physically abnormal, it was not removed, reseated, or manipulated.

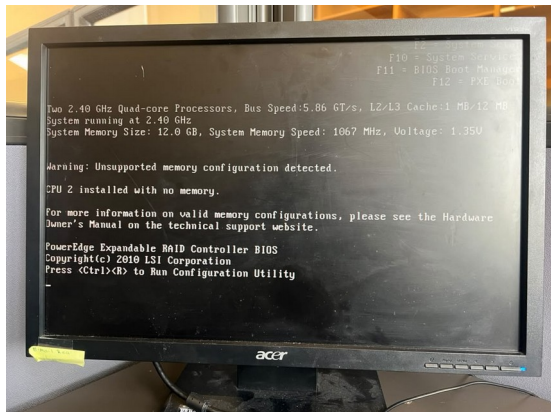
#### Other Storage / RAID Notes

- Two Western Digital Red WD10JFCX 1 TB SATA drives were mounted internally.
- A Dell FR463 3.7 V, 7 Wh RAID battery was present; battery health was not tested.
- No RAID arrays were initialized, cleared, recreated, or modified during the inspection.



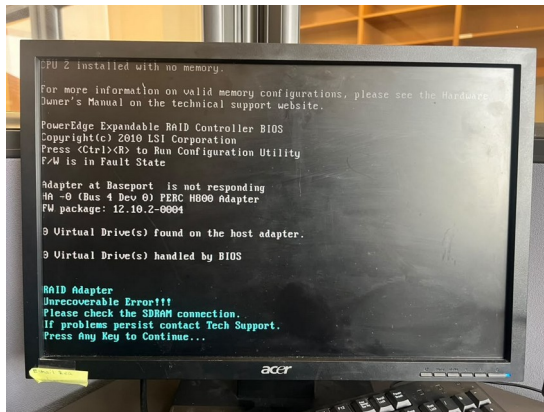
## 4. POST and BIOS Findings

### Memory Warning



POST detected both processors and 12 GB of memory, then reported: "Unsupported memory configuration detected. CPU 2 installed with no memory."

### PERC H800 Fault



POST reported a firmware fault state, an unresponsive adapter, an unrecoverable RAID error, and an instruction to check the SDRAM connection. Zero virtual drives were reported on the adapter.

### BIOS / Boot Assessment

BIOS System Setup was accessed successfully and identified the platform as a Dell PowerEdge R510 running BIOS version 1.12.0. A boot source for the operating system was then sought, but no valid bootable disk or other boot source was detected.

This outcome does not prove that the physical drives are defective. Possible causes include RAID controller initialization problems, missing or unavailable virtual-disk configuration, missing operating system data, or incorrect boot configuration.

## 5. Visual Fault Assessment

| Priority | Finding                                             | Assessment                                                  |
|----------|-----------------------------------------------------|-------------------------------------------------------------|
| High     | PERC H800 firmware fault / SDRAM connection error   | Confirmed by POST; controller did not initialize normally   |
| High     | H800-associated memory/cache module appeared arched | Physical concern; causal link not proven                    |
| Medium   | PERC H700 memory/cache module was loose             | Observed and reseated                                       |
| Medium   | CPU2 had no local memory                            | Confirmed unsupported memory population                     |
| Medium   | No bootable device detected                         | Confirmed startup outcome; cause not isolated               |
| Unknown  | Disk / array health                                 | SMART, RAID membership, and virtual-disk state not verified |

### Interpretation

The strongest remaining concern is the RAID/storage subsystem. The H800 reported a controller-level startup fault involving SDRAM, while a memory/cache module in that area appeared physically abnormal. The visual condition is relevant but does not establish a confirmed root cause. The absence of a bootable device also does not prove disk failure because RAID and boot configuration were not fully verified.

### Validation

- Server powered on and video output was restored after limited hardware intervention.
- POST detected both processors and 12 GB of installed ECC memory.
- BIOS version 1.12.0 was documented.
- PERC H800 firmware / SDRAM errors were captured.
- Installed SAS and SATA drives were physically identified.
- Components removed for documentation were returned to their original positions.

### Scope Limits

This was a physical inspection and startup assessment, not a full repair or production-readiness test. The following were not fully verified: physical drive health, RAID level and membership, virtual-disk state, usable RAID capacity, filesystem condition, operating system integrity, existing data condition, RAID battery health, network connectivity, and long-duration stability. Because more than one hardware condition changed before video output was restored, the exact cause of the original no-video condition was not isolated.

### Conclusion

The inspection established a documented inventory of the server's major hardware and captured the most important startup findings. The system can reach POST and BIOS, but it is not confirmed ready for normal service. The primary unresolved concern is the PERC H800 / SDRAM-related fault and the associated storage configuration.

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